



Flash Attention

Standard attention computes the full $N \times N$ score matrix $S = QK^T$, writes it to HBM, reads it back for softmax, writes P back, reads it again for $O = PV$.

This is memory bandwidth bound - most of the time is spent moving the data in/out of HBM, not doing the actual math.

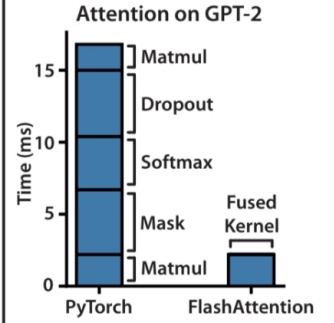
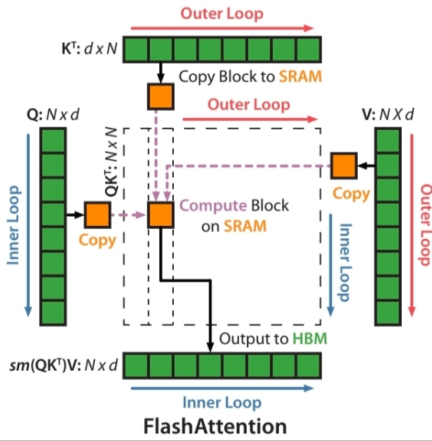
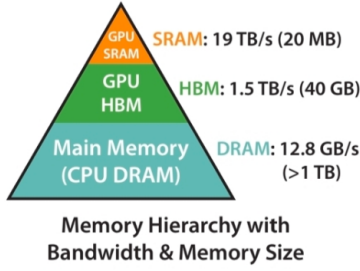
Flash Attention avoids ever materializing the full $N \times N$ matrix in HBM. It tiles Q into row blocks and K/V into column-blocks, loads only small tiles into SRAM at a time, and computes the output incrementally using online softmax - maintaining a running max (m)

- running sum (l)

- running output accumulator (O) that

gets rescaled and merged as each new K/V block is processed. By the time all column-blocks have been seen, O/I is the exact, correct softmax $(QK^T)V$ - Computed without ever storing the full S or P matrix in HBM

Result: Same output as standard attention, but less HBM reads/writes $\rightarrow$ faster and memory usage scales linearly ($O(N)$) instead of quadratically ($O(N)^2$) in sequence length.



Now, we will compute a hand drawn calculation following the exact steps below for flash Attention forward pass:

Algorithm 1 FLASHATTENTION

Require: Matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d}$ in HBM, on-chip SRAM of size M .

- 1: Set block sizes $B_c = \lceil \frac{M}{4d} \rceil$, $B_r = \min(\lceil \frac{M}{4d} \rceil, d)$.
- 2: Initialize $\mathbf{O} = (0)_{N \times d} \in \mathbb{R}^{N \times d}$, $\ell = (0)_N \in \mathbb{R}^N$, $m = (-\infty)_N \in \mathbb{R}^N$ in HBM.
- 3: Divide $\mathbf{Q}$ into $T_r = \lceil \frac{N}{B_r} \rceil$ blocks $\mathbf{Q}_1, \dots, \mathbf{Q}_{T_r}$ of size $B_r \times d$ each, and divide $\mathbf{K}, \mathbf{V}$ into $T_c = \lceil \frac{N}{B_c} \rceil$ blocks $\mathbf{K}_1, \dots, \mathbf{K}_{T_c}$ and $\mathbf{V}_1, \dots, \mathbf{V}_{T_c}$, of size $B_c \times d$ each.
- 4: Divide $\mathbf{O}$ into T_r blocks $\mathbf{O}_1, \dots, \mathbf{O}_{T_r}$ of size $B_r \times d$ each, divide ℓ into T_r blocks $\ell_1, \dots, \ell_{T_r}$ of size B_r each, divide m into T_r blocks $m_1, \dots, m_{T_r}$ of size B_r each.
- 5: **for** $1 \leq j \leq T_c$ **do**
- 6: Load $\mathbf{K}_j, \mathbf{V}_j$ from HBM to on-chip SRAM.
- 7: **for** $1 \leq i \leq T_r$ **do**
- 8: Load $\mathbf{Q}_i, \mathbf{O}_i, \ell_i, m_i$ from HBM to on-chip SRAM.
- 9: On chip, compute $\mathbf{S}_{ij} = \mathbf{Q}_i \mathbf{K}_j^T \in \mathbb{R}^{B_r \times B_c}$.
- 10: On chip, compute $\tilde{m}_{ij} = \text{rowmax}(\mathbf{S}_{ij}) \in \mathbb{R}^{B_r}$, $\tilde{\mathbf{P}}_{ij} = \exp(\mathbf{S}_{ij} - \tilde{m}_{ij}) \in \mathbb{R}^{B_r \times B_c}$ (pointwise), $\tilde{\ell}_{ij} = \text{rowsum}(\tilde{\mathbf{P}}_{ij}) \in \mathbb{R}^{B_r}$.
- 11: On chip, compute $m_i^{\text{new}} = \max(m_i, \tilde{m}_{ij}) \in \mathbb{R}^{B_r}$, $\ell_i^{\text{new}} = e^{m_i - m_i^{\text{new}}} \ell_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\ell}_{ij} \in \mathbb{R}^{B_r}$.
- 12: Write $\mathbf{O}_i \leftarrow \text{diag}(\ell_i^{\text{new}})^{-1} (\text{diag}(\ell_i) e^{m_i - m_i^{\text{new}}} \mathbf{O}_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\mathbf{P}}_{ij} \mathbf{V}_j)$ to HBM.
- 13: Write $\ell_i \leftarrow \ell_i^{\text{new}}$, $m_i \leftarrow m_i^{\text{new}}$ to HBM.
- 14: **end for**
- 15: **end for**
- 16: **Return** $\mathbf{O}$.

given one row of Q attends to $N=6$ keys,
 split into 2 blocks of 3 cols each (block size 3). Say for this row:

Block 1 scores: $S_1 = [1, 3, 2]$ $v_1 = [10, 20, 30]$

Block 2 scores: $S_2 = [5, 0, 4]$ $v_2 = [40, 50, 60]$

Please note here that we are skipping QK^T dot product.
 We basically start with step 9:

Initially, $m = -\infty$
 $l = 0$
 $o = 0$

v : marks
 value for
 local / temp
 to the current
 block

- 9: On chip, compute $S_{ij} = Q_i K_j^T \in \mathbb{R}^{B_r \times B_c}$.
 10: On chip, compute $\tilde{m}_{ij} = \text{rowmax}(S_{ij}) \in \mathbb{R}^{B_r}$, $\tilde{P}_{ij} = \exp(S_{ij} - \tilde{m}_{ij}) \in \mathbb{R}^{B_r \times B_c}$ (pointwise), $\tilde{l}_{ij} = \text{rowsum}(\tilde{P}_{ij}) \in \mathbb{R}^{B_r}$.

Inner loop

9. $S_{ij} = Q_i K_j^T \rightarrow S_1 = [1, 3, 2]$

10. $\tilde{m}_{ij} = \text{rowmax}(S_{ij}) = 3$

$\tilde{P}_{ij} = e^{(S_{ij} - \tilde{m}_{ij})} = [\exp(1-3), \exp(3-3), \exp(2-3)]$
 $= [0.1353, 1, 0.3679]$

$\tilde{l}_{ij} = \text{rowsum}(\tilde{P}_{ij}) = 0.1353 + 1 + 0.3679$
 ≈ 1.5032

Now, next step (Main step: Online softmax execution)

11: On chip, compute $m_i^{\text{new}} = \max(m_i, \tilde{m}_{ij}) \in \mathbb{R}^{B_r}$, $\ell_i^{\text{new}} = e^{m_i - m_i^{\text{new}}} \ell_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\ell}_{ij} \in \mathbb{R}^{B_r}$.

$$m_i^{\text{new}} = \max(m_i, \tilde{m}_{ij})$$

$$= \max(-\infty, 3)$$

$$= 3$$

$$\ell_i^{\text{new}} = e^{(m_i - m_i^{\text{new}})} \ell_i + e^{(\tilde{m}_{ij} - m_i^{\text{new}})} \tilde{\ell}_{ij}$$

$$= e^{(-\infty - 3)} 0 + e^{(3 - 3)} 1.5032$$

$$= e^{(-\infty)} 0 + e^0 (1.5032)$$

$$= 0 + 1.5032$$

$$\ell_i^{\text{new}} = 1.5032$$

Now,

12: Write $\mathbf{O}_i \leftarrow \text{diag}(\ell_i^{\text{new}})^{-1} (\text{diag}(\ell_i) e^{m_i - m_i^{\text{new}}} \mathbf{O}_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\mathbf{P}}_{ij} \mathbf{V}_j)$ to HBM.

$$\text{Old contribution} = \text{diag}(\ell_i) e^{m_i - m_i^{\text{new}}} \mathbf{O}_i$$

$$= \text{diag}(0) e^{(-\infty - 3)} 0$$

$$= 0$$

$$\text{New contribution} = e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{\mathbf{P}}_{ij} \mathbf{V}_j$$

$$= e^{(3 - 3)} [0.1353, 1, 0.3679] @ [10, 20, 30]$$

$$= e^0 ((0.1353 \times 10) + (1 \times 20) + (0.3679 \times 30))$$

$$= 1 (1.353 + 20 + 11.037)$$

$$= 32.39$$

$$\text{Combine:} = \text{diag}(\ell_i^{\text{new}})^{-1} (\text{old} + \text{new})$$

$$= \text{diag}(\ell_i^{\text{new}})^{-1} (0 + 32.39)$$

$$= \text{diag}(1.5032)^{-1} (32.39)$$

$$= \text{diag}\left(\frac{1}{1.5032}\right) (32.39)$$

Since this is single row, we can ignore (diag)

$$= \frac{32.39}{1.5032} \approx 21.55$$

$$O_i = 21.55$$

13: Write $\ell_i \leftarrow \ell_i^{\text{new}}$, $m_i \leftarrow m_i^{\text{new}}$ to HBM.

Now we write the values to HBM

$$\ell_i = 1.5032$$

$$m_i = 3$$

Inner loop ends

Outer loop runs again and now we go for

the next block.

$$S2 = [5, 0, 4]$$

$$V2 = [40, 50, 60]$$

step 10)

$$\tilde{m}_{ij} = \text{rowmax}(s_{ij})$$
$$= 5$$

$$\tilde{p}_{ij} = \exp(s_{ij} - \tilde{m}_{ij})$$

$$= [\exp(5-5), \exp(0-5), \exp(4-5)]$$

$$= [1, 0.0067, 0.3678]$$

$$\tilde{l}_{ij} = 1 + 0.0067 + 0.3678$$

$$\approx 1.3746$$

11)

$$m_i^{\text{new}} = \max(m_i, \tilde{m}_{ij})$$

$$= \max(3, 5)$$

$$= 5$$

$$l_i^{\text{new}} = e^{(m_i - m_i^{\text{new}})} l_i + e^{(\tilde{m}_{ij} - m_i^{\text{new}})} \tilde{l}_{ij}$$

$$= e^{(3-5)} 1.5032 + e^{(5-5)} 1.3746$$

$$= (0.1353) 1.5032 + (1)(1.3746)$$

$$= 0.2034 + 1.3746$$

$$l_i^{\text{new}} = 1.5780$$

$$12) o_i = \text{diag}(l_i^{\text{new}})^{-1} (\text{diag}(l_i) e^{m_i - m_i^{\text{new}}} o_i + e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{p}_{ij} v_j)$$

$$\text{old contribution} = \text{diag}(l_i) e^{m_i - m_i^{\text{new}}} o_i$$

$$= \text{diag}(1.5032) e^{(3-5)} (21.55)$$

$$= 1.5032 (0.1353) (21.55)$$

$$\approx 4.3829$$

$$\text{New Contribution} = e^{\tilde{m}_{ij} - m_i^{\text{new}}} \tilde{p}_{ij} v_j$$

$$\begin{aligned}
 &= e^{(5-5)} [1, 0.0067, 0.3679] @ [40, 50, 60] \\
 &= e^1 ((1 \times 40) + (0.0067 \times 50) + (0.3679 \times 60)) \\
 &= 40 + 0.337 + 22.07 \\
 &= 62.41
 \end{aligned}$$

$$\text{Combine : } \text{diag}(l_i^{\text{new}})^{-1} (4.3829 + 62.41)$$

$$= \text{diag}(1.5780)^{-1} (66.7929)$$

$$= \text{diag} \left(\frac{1}{1.5780} \right) (66.7929)$$

(ignore diag since its 1d here)

$$= \frac{66.7929}{1.5780}$$

$$O_i = 42.3275$$

Now last steps: write the l_i and m_i value to HBM along with O_i (step 12)